

EXP1:LEARNING NETWORK COMMANDS

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Course Name: Computer Networks Lab

Course Code: BCSE308L

(LINUX COMMANDS)

1.ifconfig: Shows IP address & network interfaces

```
root@Ravi-PC:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 172.22.140.124  netmask 255.255.240.0  broadcast 172.22.143.255
    inet6 fe80::215:5dff:fe58:b8a1  prefixlen 64  scopeid 0x20<link>
    ether 00:15:5d:58:b8:a1  txqueuelen 1000  (Ethernet)
    RX packets 349  bytes 335640 (335.6 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 213  bytes 17788 (17.7 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 36  bytes 5228 (5.2 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 36  bytes 5228 (5.2 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

2. mtr :

My traceroute

Hostname: google.com

1.00

—

+

Pause

Restart

About

Quit

Hostname	Loss	Snt	Last	Avg	Best	Worst	StDev
Ravi-PC.mshome.net	0.0%	12	0	0	0	0	0.1
10.110.191.156	0.0%	12	12	9	3	30	8.0
192.168.72.131	80.0%	11	36	1530	36	3023	2111.6
192.168.116.2	0.0%	11	66	63	24	99	27.7
192.168.116.33	45.5%	11	46	54	27	82	20.0
???	100.0%	11	0	0	0	0	0.0
abts-north-static-157.239.144.59.airtelbroadband.in	0.0%	11	98	73	30	127	29.9
???	100.0%	11	0	0	0	0	0.0
72.14.197.10	0.0%	11	94	69	43	101	19.5
72.14.234.9	0.0%	11	87	65	41	94	19.4
142.251.55.239	0.0%	11	47	62	39	102	18.1
hkg07s55-in-f14.1e100.net	0.0%	11	59	74	33	190	43.8

3. top: Shows running processes (live)

```
top - 03:01:55 up 7 min, 1 user, Load average: 0.00, 0.06, 0.04
Tasks: 34 total, 1 running, 33 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7788.2 total, 6767.0 free, 540.9 used, 638.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7247.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
113	root	20	0	25400	6400	4864	S	0.3	0.1	0:00.46	systemd-udevd
1	root	20	0	21676	12320	9248	S	0.0	0.2	0:02.72	systemd
2	root	20	0	3060	1664	1664	S	0.0	0.0	0:00.03	init-systemd(Ub
6	root	20	0	3076	1796	1792	S	0.0	0.0	0:00.00	init
64	root	19	-1	50436	15096	14328	S	0.0	0.2	0:00.29	systemd-journal
192	systemd+	20	0	21452	12672	10496	S	0.0	0.2	0:00.20	systemd-resolve
196	systemd+	20	0	91020	7552	6656	S	0.0	0.1	0:00.09	systemd-timesyn
213	root	20	0	4236	2432	2304	S	0.0	0.0	0:00.00	cron
215	root	20	0	6664	4096	3456	S	0.0	0.1	0:00.01	login
252	message+	20	0	9648	5120	4480	S	0.0	0.1	0:00.07	dbus-daemon
285	root	20	0	17960	8448	7552	S	0.0	0.1	0:00.08	systemd-logind
290	root	20	0	1756096	12160	10240	S	0.0	0.2	0:00.12	wsl-pro-service
296	syslog	20	0	222508	5632	4480	S	0.0	0.1	0:00.12	rsyslogd
302	root	20	0	3160	1920	1792	S	0.0	0.0	0:00.05	agetty
326	root	20	0	3116	1792	1664	S	0.0	0.0	0:00.00	agetty
327	root	20	0	107024	22656	13312	S	0.0	0.3	0:00.13	unattended-upgr
409	root	20	0	20296	11264	9216	S	0.0	0.1	0:00.14	systemd
410	root	20	0	21152	3376	1664	S	0.0	0.0	0:00.00	(sd-pam)
422	root	20	0	6072	4992	3456	S	0.0	0.1	0:00.01	bash
678	root	20	0	370076	20352	17536	S	0.0	0.3	0:00.04	packagekitd
683	polkitd	20	0	308164	7680	6912	S	0.0	0.1	0:00.05	polkitd
781	root	20	0	3064	896	896	S	0.0	0.0	0:00.00	SessionLeader
783	root	20	0	3080	1024	896	S	0.0	0.0	0:00.00	Relay(785)
785	root	20	0	6072	5248	3584	S	0.0	0.1	0:00.03	bash
886	root	20	0	3064	896	896	S	0.0	0.0	0:00.00	SessionLeader
887	root	20	0	3080	1024	896	S	0.0	0.0	0:00.00	Relay(890)
890	root	20	0	6072	5248	3584	S	0.0	0.1	0:00.03	bash
914	root	20	0	9392	4736	4352	S	0.0	0.1	0:00.02	dbus-daemon
917	root	20	0	382812	7296	6784	S	0.0	0.1	0:00.00	at-spi-bus-laun
924	root	20	0	9292	4736	4352	S	0.0	0.1	0:00.00	dbus-daemon
927	root	20	0	236068	7552	6912	S	0.0	0.1	0:00.01	at-spi2-registr

4. ps: Shows process status

```
root@Ravi-PC:~# ps -ef
```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	02:54	?	00:00:02	/sbin/init
root	2	1	0	02:54	?	00:00:00	/init
root	6	2	0	02:54	?	00:00:00	plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root	64	1	0	02:54	?	00:00:00	/usr/lib/systemd/systemd-journald
root	113	1	0	02:54	?	00:00:00	/usr/lib/systemd/systemd-udevd
systemd+	192	1	0	02:54	?	00:00:00	/usr/lib/systemd/systemd-resolved
systemd+	196	1	0	02:54	?	00:00:00	/usr/lib/systemd/systemd-timesyncd
root	213	1	0	02:54	?	00:00:00	/usr/sbin/cron -f -P
root	215	2	0	02:54	pts/1	00:00:00	/bin/login -f
message+	252	1	0	02:54	?	00:00:00	@dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-
root	285	1	0	02:54	?	00:00:00	/usr/lib/systemd/systemd-logind
root	290	1	0	02:54	?	00:00:00	/usr/libexec/wsl-pro-service -vv
syslog	296	1	0	02:54	?	00:00:00	/usr/sbin/rsyslogd -n -iNONE
root	302	1	0	02:54	hvc0	00:00:00	/sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
root	326	1	0	02:54	ttty1	00:00:00	/sbin/agetty -o -p -- \u --noclear - linux
root	327	1	0	02:54	?	00:00:00	/usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signa
root	409	1	0	02:54	?	00:00:00	/usr/lib/systemd/systemd --user
root	410	409	0	02:54	?	00:00:00	(sd-pam)
root	422	215	0	02:54	pts/1	00:00:00	-bash
root	678	1	0	02:57	?	00:00:00	/usr/libexec/packagekitd
polkitd	683	1	0	02:57	?	00:00:00	/usr/lib/polkit-1/polkitd --no-debug
root	781	2	0	02:59	?	00:00:00	/init
root	783	781	0	02:59	?	00:00:00	/init
root	785	783	0	02:59	pts/2	00:00:00	-bash
root	886	2	0	02:59	?	00:00:00	/init
root	887	886	0	02:59	?	00:00:00	/init
root	890	887	0	02:59	pts/0	00:00:00	-bash
root	914	409	0	02:59	?	00:00:00	/usr/bin/dbus-daemon --session --address=systemd: --nofork --nopidfile --systemd-activation
root	917	409	0	02:59	?	00:00:00	/usr/libexec/at-spi-bus-launcher
root	924	917	0	02:59	?	00:00:00	/usr/bin/dbus-daemon --config-file=/usr/share/defaults/at-spi2/accessibility.conf --nofork -
root	927	409	0	02:59	?	00:00:00	/usr/libexec/at-spi2-registrd --use-gnome-session
root	982	890	0	03:01	pts/0	00:00:00	top
root	990	890	0	03:03	pts/0	00:00:00	ps -ef

```
root@Ravi-PC:~#
```

5. Service network status:

In Ubuntu Linux, the **network.service** is deprecated

```
root@Ravi-PC:~# service network status
Unit network.service could not be found.
```

6. Netstat: Shows network connections

```
root@Ravi-PC:~# netstat -an
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.54:53          0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN
tcp        0      0 10.255.255.254:53      0.0.0.0:*               LISTEN
udp        0      0 127.0.0.54:53          0.0.0.0:*
udp        0      0 127.0.0.53:53          0.0.0.0:*
udp        0      0 10.255.255.254:53      0.0.0.0:*
udp        0      0 127.0.0.1:323          0.0.0.0:*
udp6       0      0 :::1:323               :::*

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix   3      [ ]         STREAM     CONNECTED    6262
unix   3      [ ]         STREAM     CONNECTED    3409
unix   2      [ ACC ]     STREAM     LISTENING    6209     /run/WSL/2_interop
unix   2      [ ACC ]     STREAM     LISTENING    10253    /run/WSL/1_interop
unix   2      [ ACC ]     SEQPACKET  LISTENING    9248     /mnt/wslg/weston-notify.sock
unix   2      [ ]         DGRAM      CONNECTED    10386
unix   2      [ ACC ]     STREAM     LISTENING    5131     /var/run/dbus/system_bus_socket
unix   3      [ ]         STREAM     CONNECTED    3555     /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED    9566
unix   2      [ ]         DGRAM      CONNECTED    12297
unix   2      [ ACC ]     STREAM     LISTENING    3394     /mnt/wslg/runtime-dir/wayland-0
unix   2      [ ACC ]     STREAM     LISTENING    3397     /tmp/.X11-unix/X0
unix   3      [ ]         STREAM     CONNECTED    2326
unix   3      [ ]         DGRAM      CONNECTED    7336
unix   3      [ ]         STREAM     CONNECTED    3717     /mnt/wslg/PulseAudioRDPSSink
unix   2      [ ACC ]     STREAM     LISTENING    7233     /mnt/wslg/runtime-dir/pulse/native
unix   2      [ ACC ]     STREAM     LISTENING    9705     /run/WSL/783_interop
unix   2      [ ACC ]     STREAM     LISTENING    2333     /mnt/wslg/PulseAudioRDPSource
```

```
unix   2      [ ACC ]     STREAM     LISTENING    9705     /run/WSL/783_interop
unix   2      [ ACC ]     STREAM     LISTENING    2333     /mnt/wslg/PulseAudioRDPSource
unix   3      [ ]         STREAM     CONNECTED    7699     /run/user/0/bus
unix   2      [ ACC ]     STREAM     LISTENING    9553     /mnt/wslg/PulseAudioRDPSSink
unix   2      [ ]         DGRAM      CONNECTED    954      /var/run/chrony/chronyd.sock
unix   3      [ ]         STREAM     CONNECTED    13445
unix   3      [ ]         STREAM     CONNECTED    9033     /tmp/dbus-6ZWoxA0Aqi
unix   3      [ ]         STREAM     CONNECTED    11390
unix   3      [ ]         DGRAM      CONNECTED    7335
unix   2      [ ]         DGRAM      CONNECTED    3664     /run/user/0/systemd/notify
unix   2      [ ACC ]     STREAM     LISTENING    3667     /run/user/0/systemd/private
unix   2      [ ACC ]     STREAM     LISTENING    10435    /run/WSL/887_interop
unix   2      [ ACC ]     STREAM     LISTENING    3675     /run/user/0/bus
unix   3      [ ]         STREAM     CONNECTED    9422
unix   3      [ ]         STREAM     CONNECTED    10373
unix   3      [ ]         STREAM     CONNECTED    8788
unix   2      [ ACC ]     STREAM     LISTENING    3676     /run/user/0/gnupg/S.dirmngr
unix   2      [ ACC ]     STREAM     LISTENING    3678     /run/user/0/gnupg/S.gpg-agent.browser
unix   2      [ ACC ]     STREAM     LISTENING    3680     /run/user/0/gnupg/S.gpg-agent.extra
unix   3      [ ]         STREAM     CONNECTED    16428
unix   2      [ ACC ]     STREAM     LISTENING    3682     /run/user/0/gnupg/S.gpg-agent
unix   2      [ ACC ]     STREAM     LISTENING    3684     /run/user/0/gnupg/S.keyboxd
unix   3      [ ]         STREAM     CONNECTED    1801
unix   2      [ ACC ]     STREAM     LISTENING    9565     /mnt/wslg/PulseServer
unix   2      [ ACC ]     STREAM     LISTENING    3686     /run/user/0/pk-debconf-socket
unix   2      [ ]         DGRAM      CONNECTED    11392
unix   3      [ ]         STREAM     CONNECTED    7695
unix   3      [ ]         STREAM     CONNECTED    7688
unix   2      [ ACC ]     STREAM     LISTENING    3688     /run/user/0/snapd-session-agent.socket
unix   2      [ ACC ]     STREAM     LISTENING    3696     /run/user/0/gnupg/S.gpg-agent.ssh
unix   2      [ ACC ]     STREAM     LISTENING    3405     /tmp/dbus-6ZWoxA0Aqi
unix   2      [ ACC ]     STREAM     LISTENING    15455    /run/systemd/resolve/io.systemd.Resolve
unix   2      [ ACC ]     STREAM     LISTENING    16893    /run/user/0/at-spi/bus
unix   2      [ ACC ]     STREAM     LISTENING    15456    /run/systemd/resolve/io.systemd.Resolve.Monitor
unix   3      [ ]         STREAM     CONNECTED    3406
unix   2      [ ACC ]     STREAM     LISTENING    3512     /run/apport.socket
unix   3      [ ]         DGRAM      CONNECTED    9295     /run/systemd/notify
unix   2      [ ACC ]     STREAM     LISTENING    3514     /run/dbus/system_bus_socket
```

7.traceroute: Shows path packets take

```
root@Ravi-PC:~# traceroute google.com
traceroute to google.com (142.251.223.238), 30 hops max, 60 byte packets
 1 Ravi-PC.mshome.net (172.22.128.1)  0.761 ms  0.730 ms  0.637 ms
 2 10.110.191.156 (10.110.191.156)  5.129 ms  5.079 ms  5.061 ms
 3 * * *
 4 192.168.116.2 (192.168.116.2)  111.876 ms  111.860 ms  111.844 ms
 5 192.168.116.49 (192.168.116.49)  110.836 ms  110.822 ms  110.809 ms
 6 * * *
 7 nsg-corporate-81.102.187.122.airtel.in (122.187.102.81)  129.752 ms  118.538 ms  118.509 ms
 8 182.79.177.69 (182.79.177.69)  133.416 ms * 116.119.161.32 (116.119.161.32)  108.839 ms
 9 72.14.197.10 (72.14.197.10)  114.105 ms  112.289 ms  113.305 ms
10 * * *
11 142.250.233.144 (142.250.233.144)  71.029 ms 142.250.233.142 (142.250.233.142)  56.476 ms 142.251.49.216 (142.251.49.216)  165.117 ms
12 142.251.55.75 (142.251.55.75)  161.134 ms 172.253.71.132 (172.253.71.132)  156.481 ms 142.251.55.75 (142.251.55.75)  155.749 ms
13 lcaaaa-al-in-f14.1e100.net (142.251.223.238)  152.794 ms  153.217 ms  108.197 ms
```

8. dig: DNS information lookup

```
root@Ravi-PC:~# dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 33716
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 87      IN      A      142.251.223.238

;; Query time: 29 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Fri Dec 19 03:10:56 UTC 2025
;; MSG SIZE rcvd: 44
```

9.host: Resolves domain -IP

```
root@Ravi-PC:~# host google.com
google.com has address 142.251.223.238
google.com has IPv6 address 2404:6800:4007:832::200e
google.com mail is handled by 10 smtp.google.com.
```

10.hostname: Shows system name

```
root@Ravi-PC:~# hostname
Ravi-PC
```

11. hostid: Shows unique host identifier

```
root@Ravi-PC:~# hostid
007f0101
```

12. Ping: Checks network connectivity

```
root@Ravi-PC:~# ping google.com
PING google.com (172.217.24.206) 56(84) bytes of data.
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=1 ttl=114 time=192 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=2 ttl=114 time=63.1 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=3 ttl=114 time=98.8 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=4 ttl=114 time=53.6 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=5 ttl=114 time=86.8 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=6 ttl=114 time=50.0 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=7 ttl=114 time=80.4 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=8 ttl=114 time=54.3 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=9 ttl=114 time=74.3 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=10 ttl=114 time=97.5 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=11 ttl=114 time=55.1 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=12 ttl=114 time=89.4 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=13 ttl=114 time=47.2 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=14 ttl=114 time=76.1 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=15 ttl=114 time=173 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=16 ttl=114 time=83.2 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=17 ttl=114 time=40.0 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=18 ttl=114 time=93.4 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=19 ttl=114 time=86.4 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=20 ttl=114 time=66.0 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=21 ttl=114 time=131 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=22 ttl=114 time=55.6 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=23 ttl=114 time=89.9 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=24 ttl=114 time=50.2 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=25 ttl=114 time=83.2 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=26 ttl=114 time=60.2 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=27 ttl=114 time=69.1 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=28 ttl=114 time=114 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=29 ttl=114 time=63.0 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=30 ttl=114 time=98.5 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=31 ttl=114 time=56.8 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=32 ttl=114 time=89.9 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=33 ttl=114 time=48.7 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=34 ttl=114 time=81.8 ms
64 bytes from hkg12s13-in-f14.1e100.net (172.217.24.206): icmp_seq=35 ttl=114 time=200 ms
```

13. arp: Shows ARP table

```
root@Ravi-PC:~# arp -a
Ravi-PC.mshome.net (172.22.128.1) at 00:15:5d:08:16:be [ether] on eth0
```

14. ss: Shows socket statistics (modern netstat)

```
root@Ravi-PC:~# ss -tuln
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
udp	UNCONN	0	0	127.0.0.54:53	0.0.0.0:*	
udp	UNCONN	0	0	127.0.0.53%lo:53	0.0.0.0:*	
udp	UNCONN	0	0	10.255.255.254:53	0.0.0.0:*	
udp	UNCONN	0	0	127.0.0.1:323	0.0.0.0:*	
udp	UNCONN	0	0	:::1:323	:::1:*	
tcp	LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*	
tcp	LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*	
tcp	LISTEN	0	1000	10.255.255.254:53	0.0.0.0:*	

15. nslookup: DNS lookup

```
root@Ravi-PC:~# nslookup google.com
Server:          10.255.255.254
Address:         10.255.255.254#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.24.206
Name:   google.com
Address: 2404:6800:4007:835::200e
```

16. route: Shows routing table

```
root@Ravi-PC:~# route
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
default        Ravi-PC.mshome. 0.0.0.0         UG    0      0      0 eth0
172.22.128.0   0.0.0.0         255.255.240.0   U     0      0      0 eth0
```

(WINDOWS COMMANDS)

1. ipconfig:

```
C:\Users\ravib>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40f4:40d5:47aa:6697:70d5:5707:d053
    Temporary IPv6 Address. . . . . : 2409:40f4:40d5:47aa:41b3:7b7c:97fd:26a3
    Link-local IPv6 Address . . . . . : fe80::24f8:72f3:38c7:b57d%10
    IPv4 Address. . . . . : 10.138.35.47
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::20ea:a3ff:fe60:1a49%10
                                10.138.35.125

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::299:e2cf:ce7f:d440%38
    IPv4 Address. . . . . : 172.22.128.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :
```

2. pathping:

```
C:\Users\ravib>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
               [-p period] [-q num_queries] [-w timeout]
               [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period          Wait period milliseconds between pings.
    -q num_queries     Number of queries per hop.
    -w timeout         Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```


3. TaskManager:

Processes

Run new task

End task

Efficiency mode

...

Name	Status	12% CPU	79% Memory	0% Disk	0% Network
Apps (1)					
> Task Manager		0%	70.6 MB	0 MB/s	0 Mbps
Background processes (85)					
> 64-bit Synaptics Pointing Enha...		0%	1.2 MB	0 MB/s	0 Mbps
ApplicationFrameHost		0%	3.2 MB	0 MB/s	0 Mbps
BraveSoftware Update (32 bit)		0%	0.5 MB	0 MB/s	0 Mbps
chrome	Efficiency ...	0%	10.6 MB	0 MB/s	0 Mbps
chrome	Efficiency ...	0%	10.9 MB	0 MB/s	0 Mbps
chrome		0%	1.6 MB	0 MB/s	0 Mbps

4.Tasklist:

Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	136 K
Secure System	236	Services	0	66,172 K
Registry	280	Services	0	39,808 K
smss.exe	724	Services	0	712 K
csrss.exe	752	Services	0	3,752 K
wininit.exe	1084	Services	0	4,044 K
services.exe	1160	Services	0	9,324 K
LsaIso.exe	1180	Services	0	1,768 K
lsass.exe	1188	Services	0	25,572 K
fontdrvhost.exe	1364	Services	0	944 K
svchost.exe	1440	Services	0	63,996 K
WUDFHost.exe	1500	Services	0	11,988 K
svchost.exe	1588	Services	0	24,788 K
svchost.exe	1632	Services	0	8,468 K
WUDFHost.exe	1724	Services	0	5,744 K
svchost.exe	1860	Services	0	1,556 K
svchost.exe	1880	Services	0	3,268 K
svchost.exe	1892	Services	0	10,796 K
svchost.exe	2044	Services	0	7,080 K
WUDFHost.exe	1316	Services	0	2,596 K
svchost.exe	1296	Services	0	12,608 K
svchost.exe	2008	Services	0	5,276 K

5. net start:

```
C:\Users\ravib>net start
These Windows services are started:

Application Information
AppX Deployment Service (AppXSVC)
AVCTP service
Background Intelligent Transfer Service
Background Tasks Infrastructure Service
Base Filtering Engine
BitLocker Drive Encryption Service
Bluetooth Audio Gateway Service
Bluetooth Support Service
Capability Access Manager Service
Client License Service (ClipSVC)
Clipboard User Service_3f3834cc
CNG Key Isolation
COM+ Event System
Connected Devices Platform Service
Connected Devices Platform User Service_3f3834cc
Connected User Experiences and Telemetry
CoreMessaging
Credential Manager
Cryptographic Services
Data Usage
DCOM Server Process Launcher
Delivery Optimization
Device Association Service
DevicesFlow_3f3834cc
DHCP Client
Diagnostic Policy Service
Diagnostic Service Host
Diagnostic System Host
Display Enhancement Service
Display Policy Service
Distributed Link Tracking Client
DNS Client
DtsApo4Service
Extensible Authentication Protocol
FileSyncHelper
Geolocation Service
Group Policy Client
```

6. tracert:

```
C:\Users\ravib> traceert
'traceert' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\ravib>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                Do not resolve addresses to hostnames.
    -h maximum_hops  Maximum number of hops to search for target.
    -j host-list      Loose source route along host-list (IPv4-only).
    -w timeout        Wait timeout milliseconds for each reply.
    -R                Trace round-trip path (IPv6-only).
    -S srcaddr        Source address to use (IPv6-only).
    -4                Force using IPv4.
    -6                Force using IPv6.
```

7. dig = nslookup:

```
C:\Users\ravib>nslookup
Default Server: UnKnown
Address: 10.138.35.125

> google.com
Server: UnKnown
Address: 10.138.35.125

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4002:802::200e
          142.250.182.142
```

8. host=nslookup

```
> nslookup
Server: UnKnown
Address: 10.138.35.125
```

9. hostname:

```
C:\Users\ravib>hostname
Ravi-PC
```

10. wmic csproduct get uuid:

```
C:\Users\ravib>wmic csproduct get uuid
UUID
AED26B41-F729-EF11-A4F6-28C5C82DBDB2
```

11. ping:

```
C:\Users\ravib>ping google.com

Pinging google.com [2404:6800:4002:822::200e] with 32 bytes of data:
Reply from 2404:6800:4002:822::200e: time=113ms
Reply from 2404:6800:4002:822::200e: time=334ms
Reply from 2404:6800:4002:822::200e: time=64ms
Reply from 2404:6800:4002:822::200e: time=289ms

Ping statistics for 2404:6800:4002:822::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 64ms, Maximum = 334ms, Average = 200ms
```

12. arp:

```
Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

    -a                Displays current ARP entries by interrogating the current
                        protocol data. If inet_addr is specified, the IP and Physical
                        addresses for only the specified computer are displayed. If
                        more than one network interface uses ARP, entries for each ARP
                        table are displayed.
    -g                Same as -a.
    -v                Displays current ARP entries in verbose mode. All invalid
                        entries and entries on the loop-back interface will be shown.
    inet_addr          Specifies an internet address.
    -N if_addr        Displays the ARP entries for the network interface specified
                        by if_addr.
    -d                Deletes the host specified by inet_addr. inet_addr may be
                        wildcarded with * to delete all hosts.
    -s                Adds the host and associates the Internet address inet_addr
                        with the Physical address eth_addr. The Physical address is
                        given as 6 hexadecimal bytes separated by hyphens. The entry
                        is permanent.
    eth_addr           Specifies a physical address.
    if_addr            If present, this specifies the Internet address of the
                        interface whose address translation table should be modified.
                        If not present, the first applicable interface will be used.

Example:
> arp -s 157.55.85.212 00-aa-00-62-c6-09 .... Adds a static entry.
> arp -a              .... Displays the arp table.
```

13. netstat:

```
C:\Users\ravib>netstat
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	10.138.35.47:50630	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:50690	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:50723	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:51804	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:52589	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:52590	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:53430	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:53692	20.195.84.7:https	ESTABLISHED
TCP	10.138.35.47:54223	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:54244	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:54572	10.138.35.125:domain	TIME_WAIT
TCP	10.138.35.47:56304	10.138.35.125:domain	TIME_WAIT

14. nslookup:

```
C:\Users\ravib>nslookup
Default Server:  UnKnown
Address:  10.138.35.125
```

```
> google.com
Server:  UnKnown
Address:  10.138.35.125
```

```
Non-authoritative answer:
Name:      google.com
Addresses: 2404:6800:4002:802::200e
           142.250.182.142
```

15. route print:

```
C:\Users\ravib>route print
=====
Interface List
17...28 c5 c8 2d bd b2 .....Realtek Gaming GbE Family Controller
6...dc 45 46 06 4e 75 .....Microsoft Wi-Fi Direct Virtual Adapter
16...de 45 46 06 4e 74 .....Microsoft Wi-Fi Direct Virtual Adapter #2
10...dc 45 46 06 4e 74 .....Intel(R) Wi-Fi 6E AX211 160MHz
1.....Software Loopback Interface 1
38...00 15 5d 08 16 be .....Hyper-V Virtual Ethernet Adapter
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          10.138.35.125    10.138.35.47     50
10.138.35.0                255.255.255.0    On-link          10.138.35.47     306
10.138.35.47               255.255.255.255  On-link          10.138.35.47     306
10.138.35.255              255.255.255.255  On-link          10.138.35.47     306
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
172.22.128.0               255.255.240.0    On-link          172.22.128.1     5256
172.22.128.1               255.255.255.255  On-link          172.22.128.1     5256
172.22.143.255             255.255.255.255  On-link          172.22.128.1     5256
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link          10.138.35.47     306
224.0.0.0                  240.0.0.0        On-link          172.22.128.1     5256
255.255.255.255            255.255.255.255  On-link          127.0.0.1        331
255.255.255.255            255.255.255.255  On-link          10.138.35.47     306
255.255.255.255            255.255.255.255  On-link          172.22.128.1     5256
=====
Persistent Routes:
Network Address            Netmask          Gateway Address  Metric
0.0.0.0                    0.0.0.0          192.168.0.1     Default
```